

Week 4 Capstone Project

End-to-End Sales Analysis and Forecasting

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Course: Data Science with Python

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Data Visualization & Capstone Project

This report presents data cleaning, exploratory data analysis, sales prediction using regression, Excel analysis, Power BI dashboard requirements, findings and recommendations for the provided sales dataset.

1. Dataset Overview

The dataset contains 700 records and 16 columns. The dates range from September 2013 to December 2014. There are no duplicate rows after cleaning. The Discount Band column had 53 missing values, which were treated as 'Unknown'.

2. Key Metrics

Metric	Value
Total Sales	118,726,350.26
Total Profit	16,893,702.26
Total Units Sold	1,125,806
Gross Sales	127,931,598.50
Discounts	9,205,248.24
Overall Profit Margin	14.23%

3. Exploratory Data Analysis

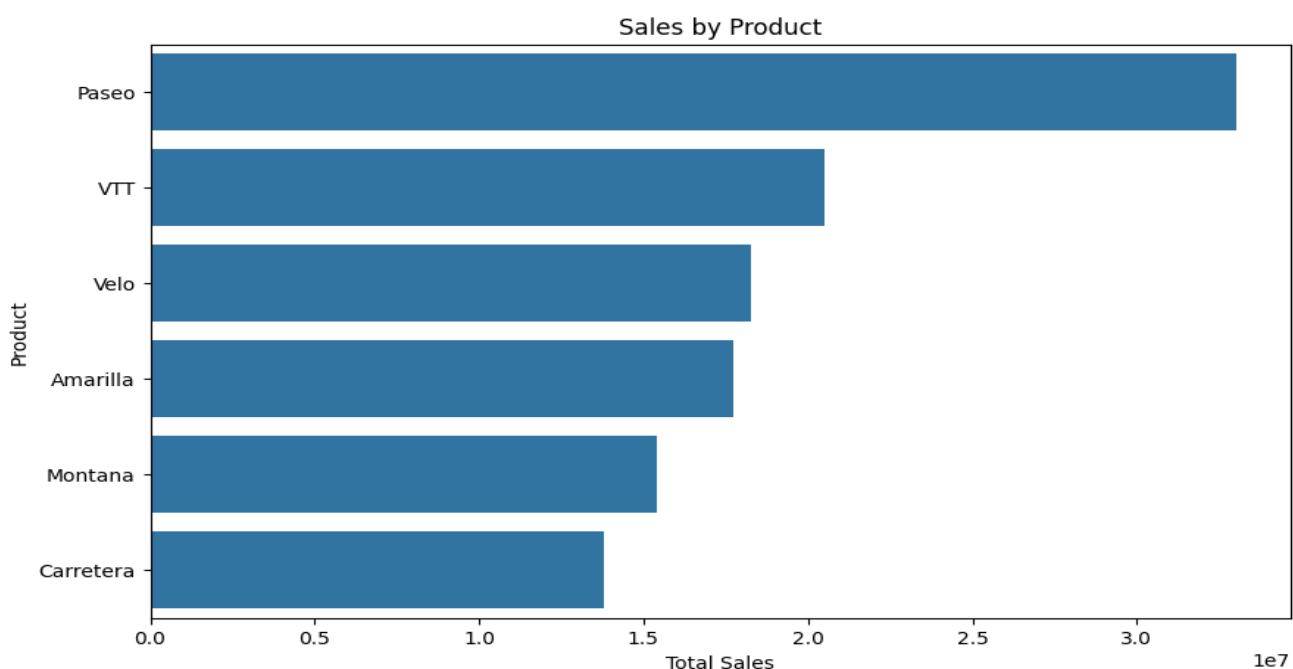


Fig. Sales by Product

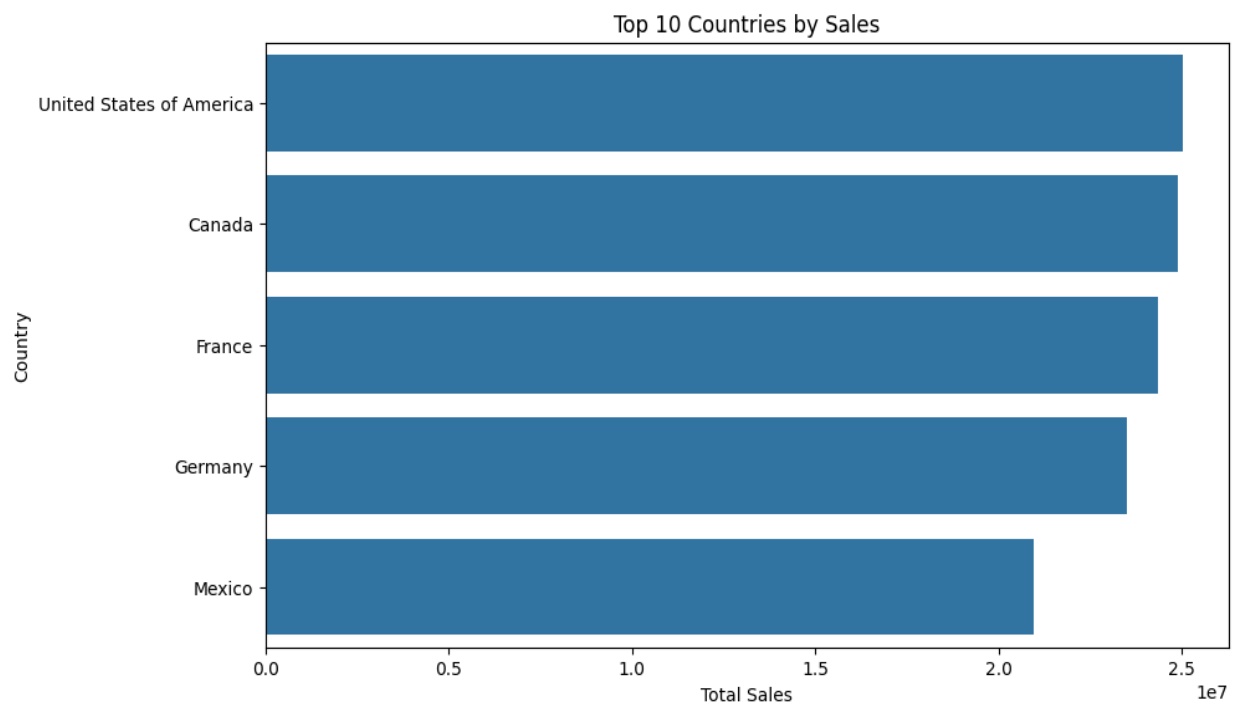


Fig. Sales by Country

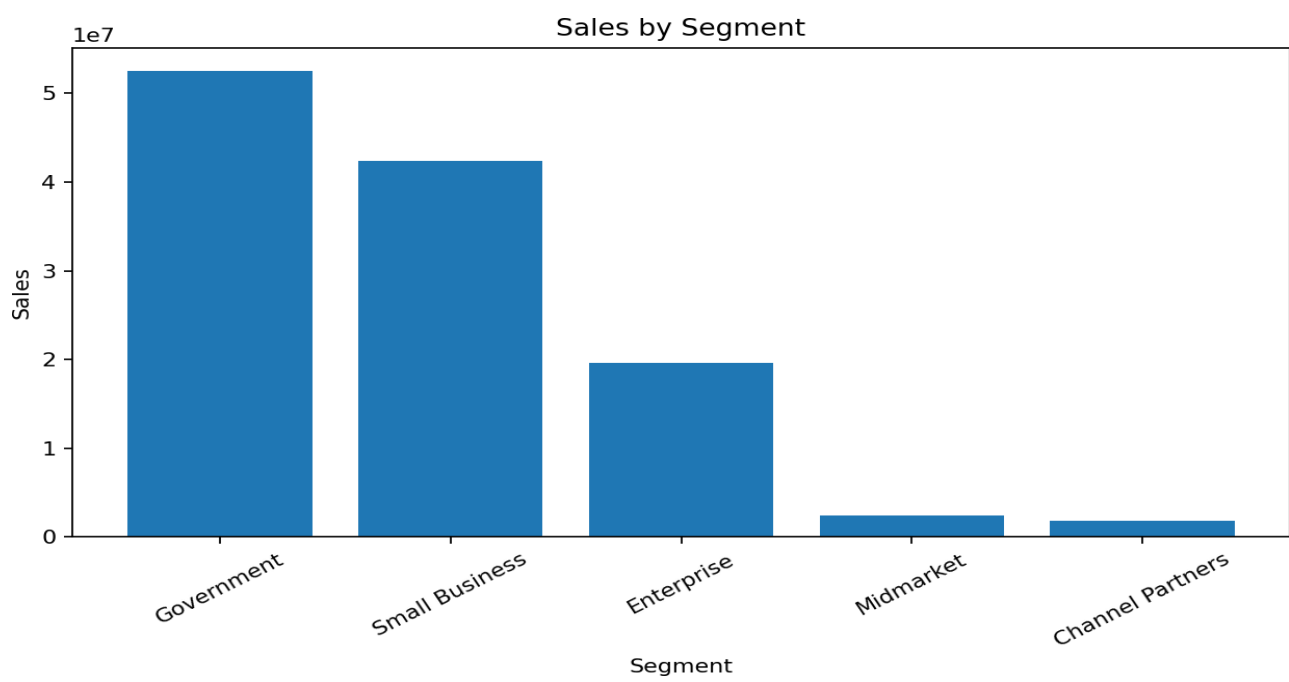


Fig. Sales vs Units Sold

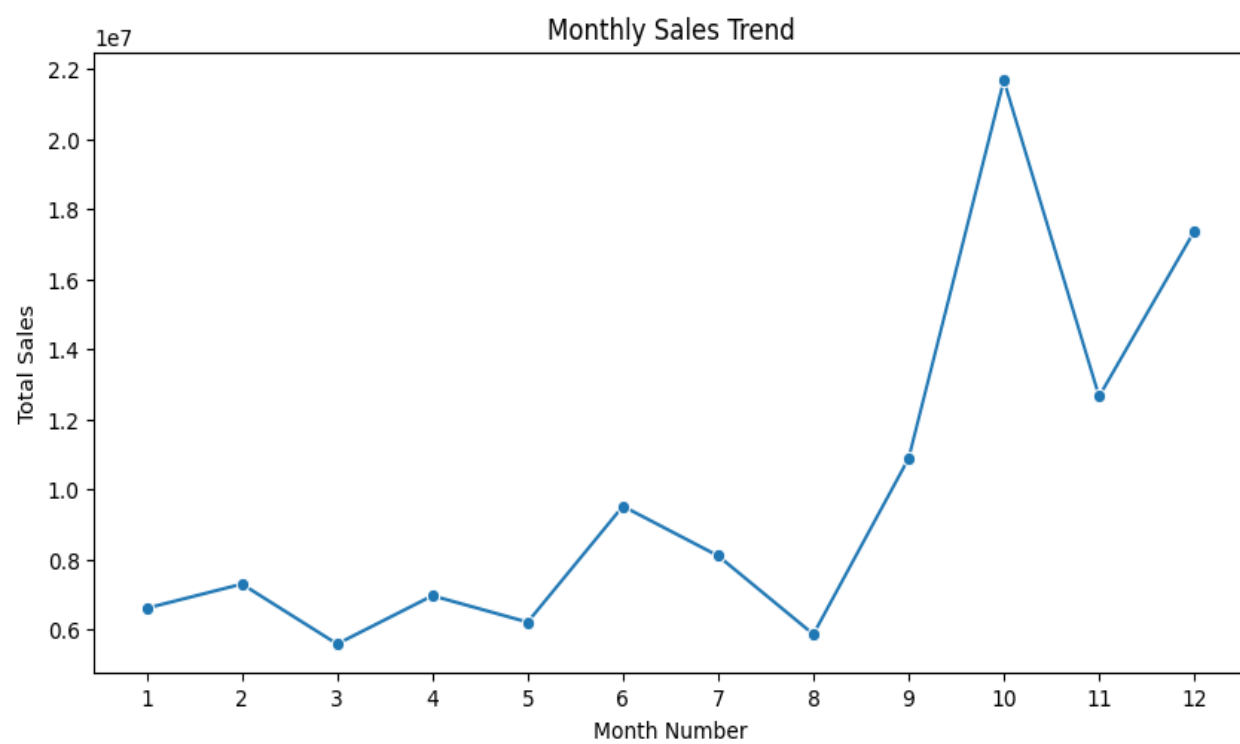


Fig. Monthly Sales Trend

4. Regression Model

A Random Forest Regression model was used to predict Sales. The input variables included segment, country, product, discount band, units sold, manufacturing price, sale price, month number and year. Gross Sales, Discounts, COGS and Profit were not used as predictors because they are directly related to the Sales calculation and could cause target leakage.

Metric	Result
Model	Random Forest Regressor
Training Records	560
Testing Records	140
MAE	10,904.44
RMSE	29,112.95
R ² Score	0.9847

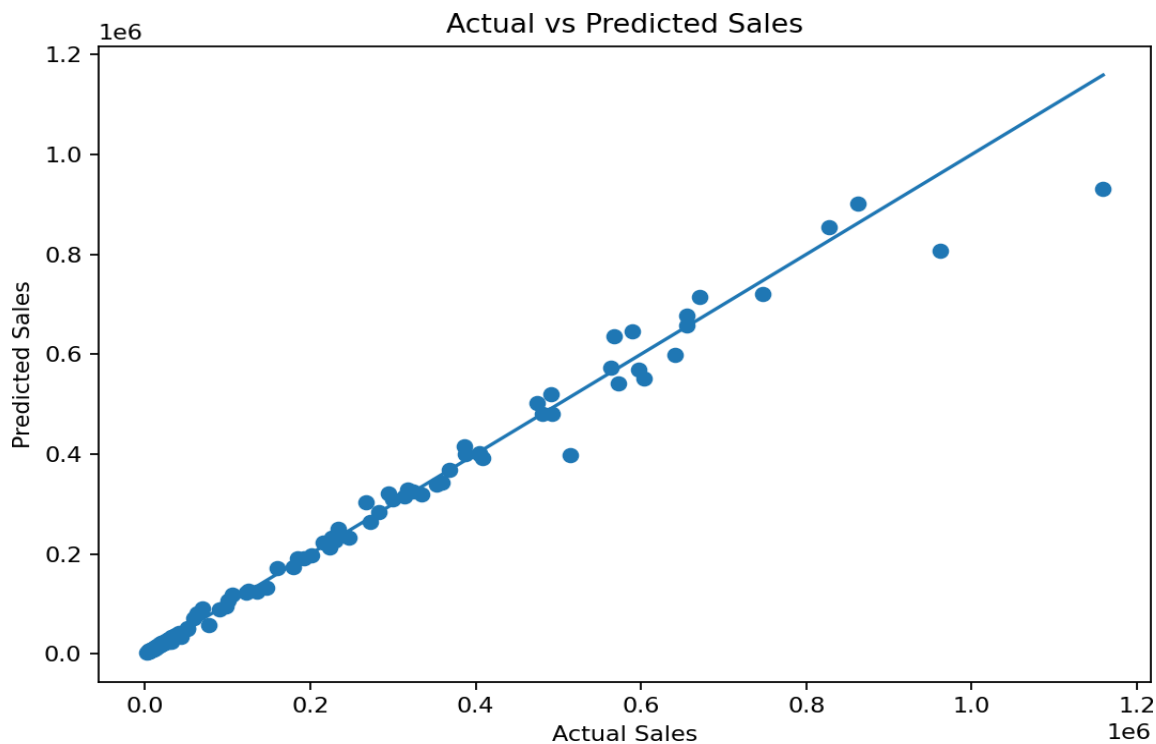


Fig. Actual versus predicted sales on the test set.

5. Key Findings

- Total sales are 118.73 million and total profit is 16.89 million.
- Paseo is the highest-sales product with 33.01 million in sales.
- Government is the largest sales segment with 52.50 million in sales.
- Enterprise records negative total profit of 614,545.62 in the dataset.
- 2014 has substantially higher sales than 2013.
- The low discount band has a higher profit margin than the high discount band.

6. Recommendations

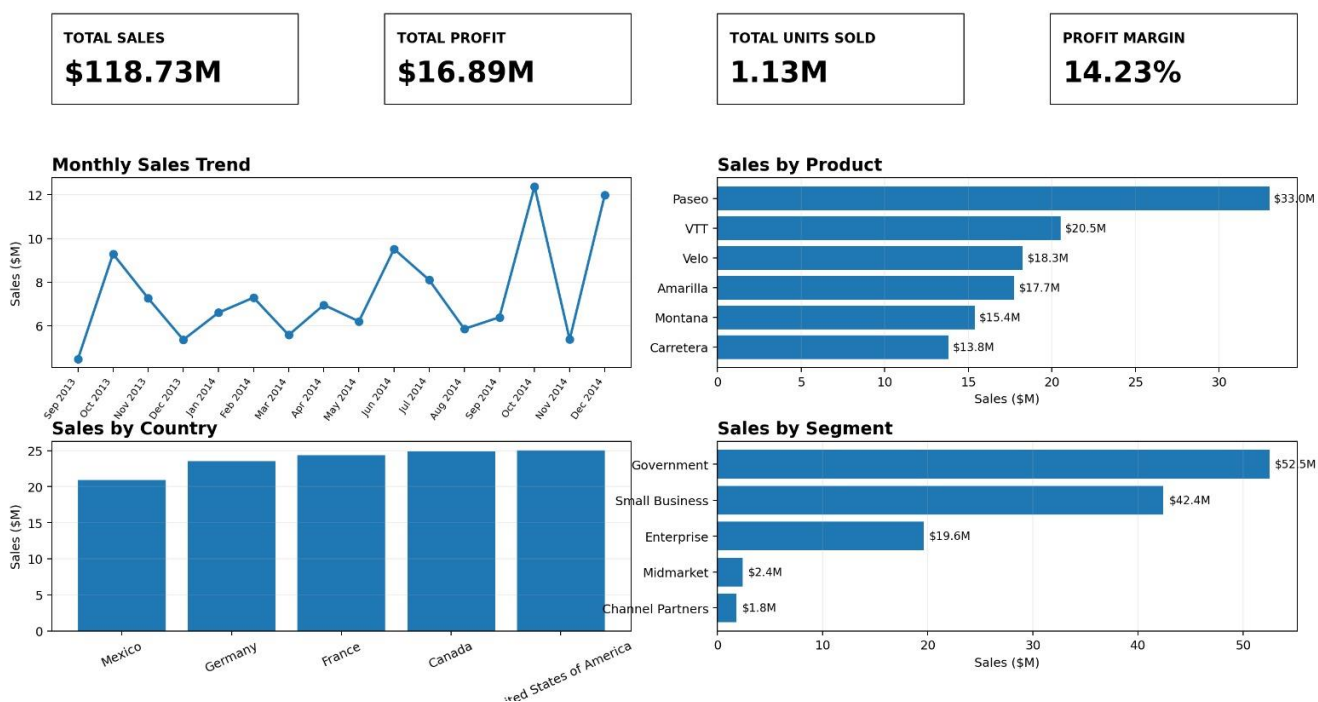
- Monitor high-sales products and segments along with their profit margins.
- Investigate the Enterprise segment to understand the reasons for its negative total profit.
- Use monthly sales trends for inventory and sales planning.
- Review discount levels because higher discounts are associated with lower profit margin in this dataset.
- Use the regression model as a supporting forecasting tool and validate predictions with future business data.

7. Power BI Dashboard

The Power BI dashboard should contain Total Sales, Total Profit, Total Units Sold and Profit Margin KPI cards; Sales by Product, Sales by Country, Sales by Segment and Monthly Sales Trend charts; and slicers for Year, Country, Product, Segment and Discount Band. The cleaned Excel workbook supplied with this project can be used as the Power BI data source.

SALES & PROFIT PERFORMANCE DASHBOARD

Business Performance Analysis • 2013–2014



Conclusion

This project successfully completed data cleaning, exploratory data analysis, sales prediction using machine learning, and dashboard planning. The analysis identified key sales drivers, profitable segments, and opportunities for business improvement. The regression model achieved strong predictive performance and can support future sales forecasting.

GitHub Repository Link

<https://github.com/sabalegayatri23/skillnexus-data-science-internship>

